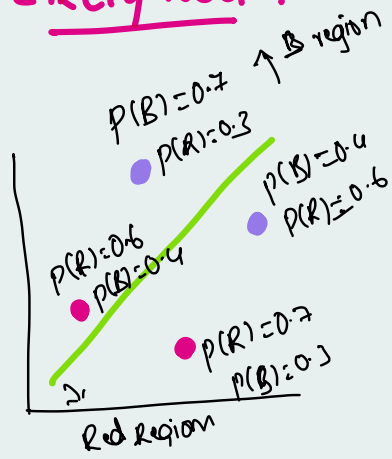
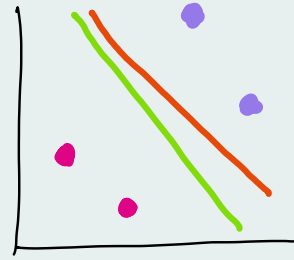


Maximum Likelihood :



model - 1

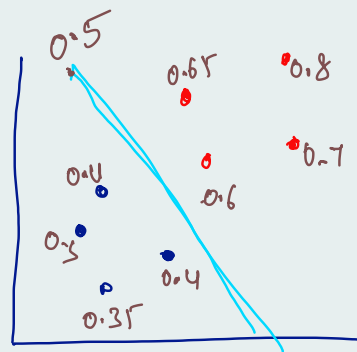
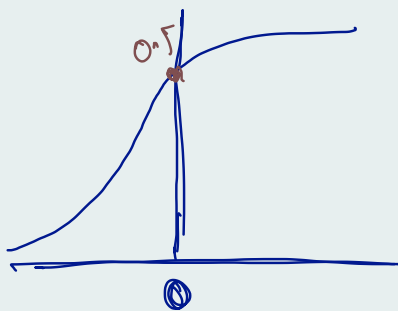


model - 2

— out of 2 model which model is best ?

— Obviously model 2. But

If one more line (orange) is added then which is best fit line. This can be identified by the loss function which is Maximum likelihood



$$\sigma = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\hat{y} = \sigma(z)$$

$$z = \sum w_i x_i$$

$$\text{model 1} = 0.7 \times 0.6 \times 0.4 \times 0.7$$

$$\text{model 2} = 0.7 \times 0.6 \times 0.6 \times 0.7$$

which ever the value is maximum will consider as the best fit line.

- If the no. of points increases multiplication value will become zero ($\approx$)

- So instead let's apply addition

- But for this addition will use 'log'

$$\Rightarrow \log(ab) = \log(a) + \log(b)$$

$$\log(\text{max}) = \log(0.7) + \log(0.6) + \log(0.4) + \log(0.7)$$

[Note: 0-1 range value will be -ve values in log]

$$\text{Ex: } \log(0.9) = -0.045$$

$$\therefore \log(\text{max}) = -\log(0.7) - \log(0.6) - \log(0.4) - \log(0.7)$$

↓
now this equation call it as

Cross Entropy.

Cross Entropy is summation of maximum likelihood log values.

Maximum likelihood $\uparrow$ But Cross Entropy $\downarrow$

$$\text{Bcz } \log(0.1) > \log(0.9)$$

$\downarrow$ $\downarrow$
 -1 -0.04

— Now whichever the model C.E value is low will be best model.

Equation Simplification

$$\log(\max) = -\log(\hat{y}_1) - \log(\hat{y}_2) - \log(\hat{y}_3) - \log(\hat{y}_4)$$

— But it will not work well with misclassification
so ~~for~~ we should include both the classes.

$$[-y_i \log(\hat{y}_i) - (1-y_i) \log(1-\hat{y}_i)]$$

Testing: If $y_i = 1, \hat{y}_i = 0.7$

$$\Rightarrow -1 \log(0.7) - (1-1) \log(1-0.7)$$

$$\Rightarrow -\log(0.7) - 0 = -\log(0.7)$$

② If $y_i = 0, \hat{y}_i = 0.6$

$$\Rightarrow 0 \cdot \log(0.6) - (1-0) \log(1-0.6)$$

$$\Rightarrow -\log(0.4)$$

Note: These values are equal to previous equation

$$\log(\max) = -\log(0.7) - \log(0.4) \dots$$

Final Loss Function is

$$L = \sum_{i=1}^n -y_i \log(\hat{y}_i) - (1-y_i) \log(1-\hat{y}_i)$$

— we are calling this as log-loss Error
and Binary Cross Entropy.

— with this we are trying to minimize the loss.
and finding w_1, w_2, w_0 for best fit line.

— But for finding these w_1, w_2, w_0 no close form
equation is available like linear Reg (OLS)

— we are going to use Gradient Descent to
find these coeff.

Logistic Reg :

Linear Regr equation is

output $\hat{y} = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots$

coeff/wts

features

Range of Linear Reg is $[-\infty \text{ to } +\infty]$

and the coeff of Linear Reg $\beta_1, \beta_2 \dots$ are continuous

range is $[-\infty \text{ to } +\infty]$

$$\therefore (-\infty, +\infty) = (-\infty, +\infty)$$

- When we apply the same eqn on Logit Reg which is categorical in nature $\{0, 1\}$

- So the range is mismatch

$$0/1 \neq (-\infty, +\infty)$$

- We need to make the ranges are equal.

- So first let's apply probability.

$$p(y=1) = \beta_0 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots$$

$$[0, 1] \neq (-\infty, \infty)$$

- Still after applying prob the ranges are not matching

- So prob is not correct replacement.

- So let's introduce one more concept called

Odds ratio.

$$\text{Odds} = \frac{p}{1-p} = \frac{\text{prob of event occur}}{\text{prob of event not occur.}}$$

Ex: what is the prob of getting 1 in dice.

$$p = \frac{1}{6}, \quad \text{Odds} = \frac{1}{5} = \frac{\frac{1/6}{1-1/6}}{\frac{1/6}{5/6}} = \frac{1}{5}$$

Note: The range of odds is $[0, \infty)$

so let's replace in equ.

$$\frac{p}{1-p} = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots$$

$$[0, \infty) \neq (-\infty, +\infty)$$

- let's apply 'log'

$$\log\left(\frac{p}{1-p}\right) = \beta_1 x_1 + \beta_2 x_2 + \dots$$

- Note: Log of any decimal values are -ve value.

$$\log(0.5) = -0.3010$$

$$\log(5) = 0.698$$

so log of any numbers are ranging from $(-\infty, \infty)$

$$\therefore (-\infty, \infty) = (-\infty, +\infty)$$

Now the ranges are matching.

$$\boxed{\therefore \log\left(\frac{p}{1-p}\right) = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots} \rightarrow \text{equ of LOR.}$$

Though LOR is used for classification, still we calling it as regression,

Reason is equ is derived from LR equ.

we are calling $\log\left(\frac{p}{1-p}\right)$ as logit function.

— In LR, we regressed/predicted on target column ($\hat{y}$)

— But in LOR, we regressed on logit fun, that is why we are calling LOR.

— Now in order predict 0/1, we should take prob

$$p = \frac{1}{1 + e^{-(\beta_1 x_1 + \beta_2 x_2 + \dots)}} \quad \begin{array}{l} \text{range is } (-\infty \text{ to } \infty) \\ \downarrow \\ \text{how to bring } [0, 1] \end{array}$$

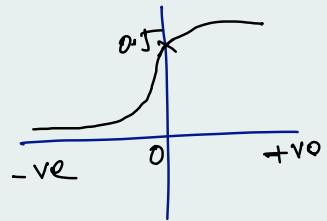
This is the equation of sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

— For any big -ve value, $\sigma(-ve) \approx 0$

— For any big +ve value, $\sigma(+ve) \approx 1$

$$\sigma(0) = 0.5$$



$$\sigma(0) = \frac{1}{1 + e^0} = \frac{1}{1+1} = 0.5$$

— we are going to apply LOR on dataset and will get probability values as output.

— Loss Function of LR is MSE

- Loss fun of LOR is

logloss (categorical cross entropy)

actual prob val
 y_i p

$$\text{logloss} = - \sum_{i=1}^N (y_i \log p + (1-y_i) \log(1-p))$$

entropy measures uncertainty in the data.

$$= -1 \cdot \log(0.2) + (1-1) \log(1-0.2)$$

$$= -\log(0.2) + 0$$

$$= -(-0.69) = 0.69 \text{ is loss.}$$

y_i	p
1	0.2
1	0.9
0	0.8

$$2. \quad = -1 \log(0.9) + 0$$

$$= -(-0.04) = 0.04.$$

$$3. \quad = -0 \cdot \log(0.8) + (1-0) \log(1-0.8)$$

$$= 0 + \log(0.2) = -0.698.$$

